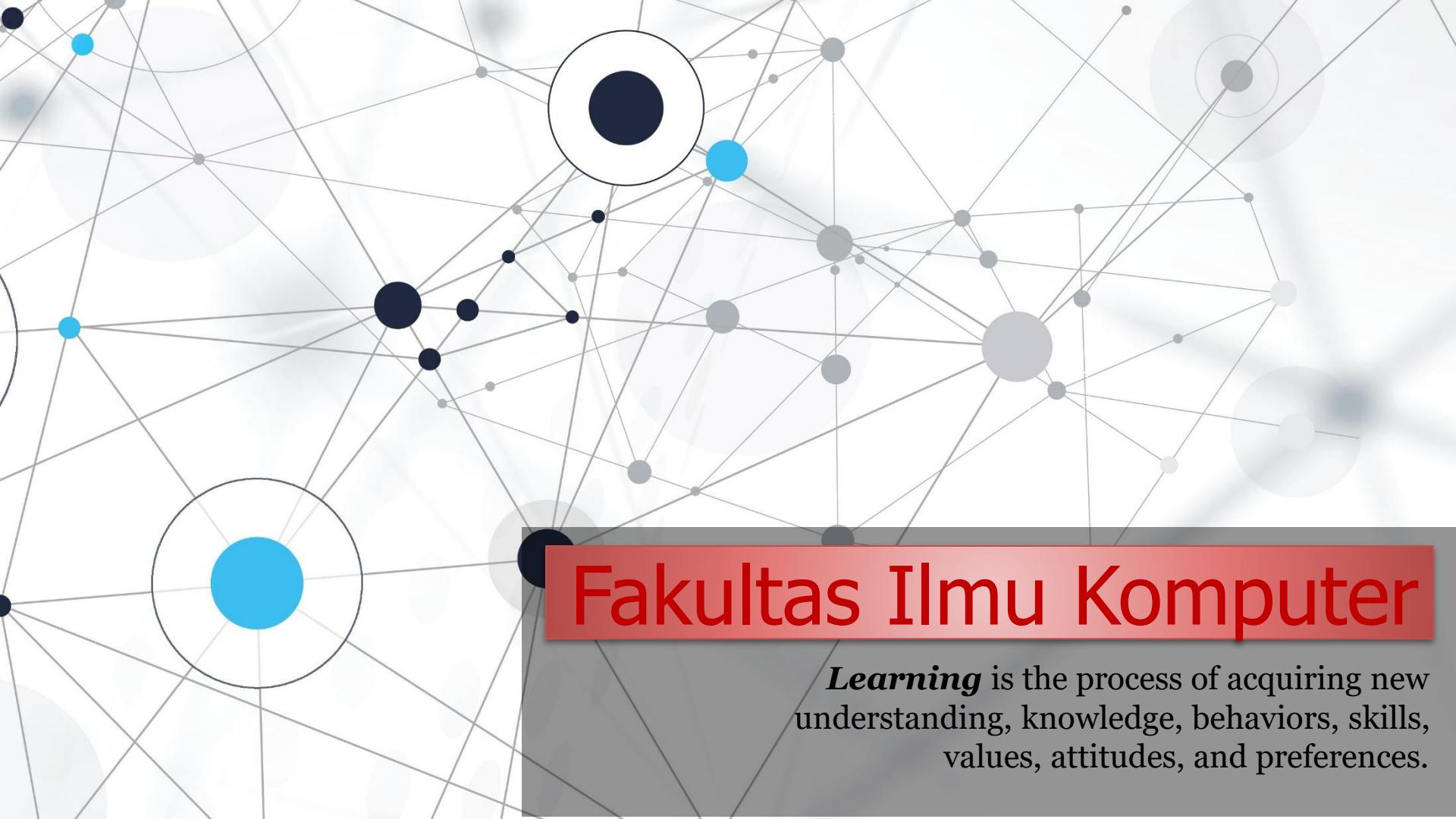




Fakultas Ilmu Komputer

Learning is the process of acquiring new understanding, knowledge, behaviors, skills, values, attitudes, and preferences.

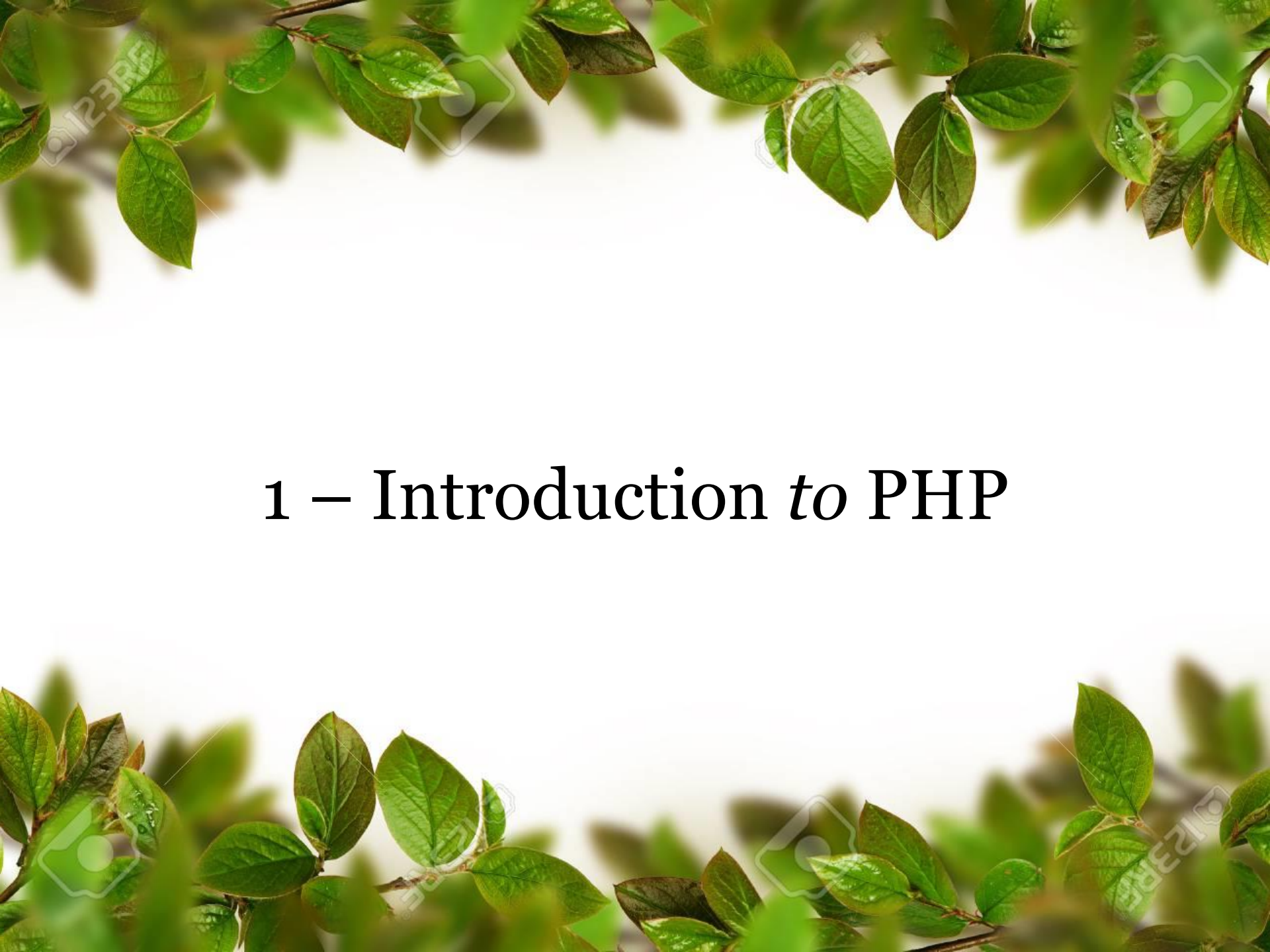
Server-side Scripting Language

[*PHP Dynamic Web Development*]



Learn PHP *for* Beginners

PHP (recursive acronym for PHP: Hypertext Preprocessor)

The slide features a white background with green leaves and branches framing the top and bottom edges. The leaves are vibrant green and appear to be from a tree or shrub. The text is centered in the middle of the slide.

1 – Introduction *to* PHP

Full Stack Development Concept

**Sebutkan hewan-hewan yang
bisa bertahan hidup *di* darat, air
dan udara?**

(What are the animals that can live on land, water and air?)



Kapal Angkatan Laut
(Navy Ship)



Pesawat Angkatan Udara
(Air Force Aircraft)



Tank Angkatan Darat
(Army Tanks)

Full Stack Development Concept



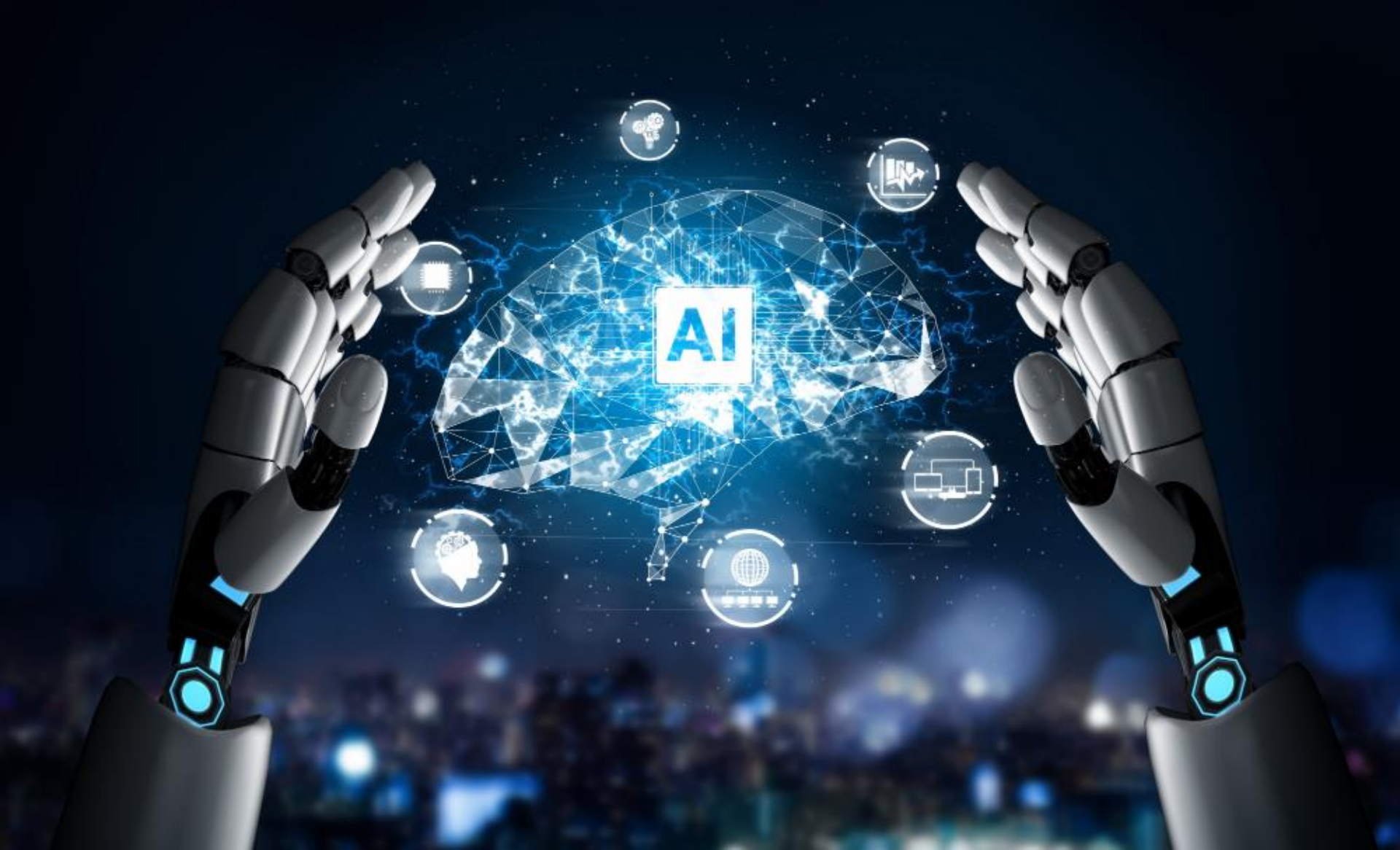
Di darat, dapatkah bebek
berlari lebih cepat dari seekor
anjing?



Di laut, dapatkah bebek
berenang lebih cepat dari
seekor lumba-lumba?



Di udara, dapatkah bebek terbang
lebih cepat dari seekor burung
elang?

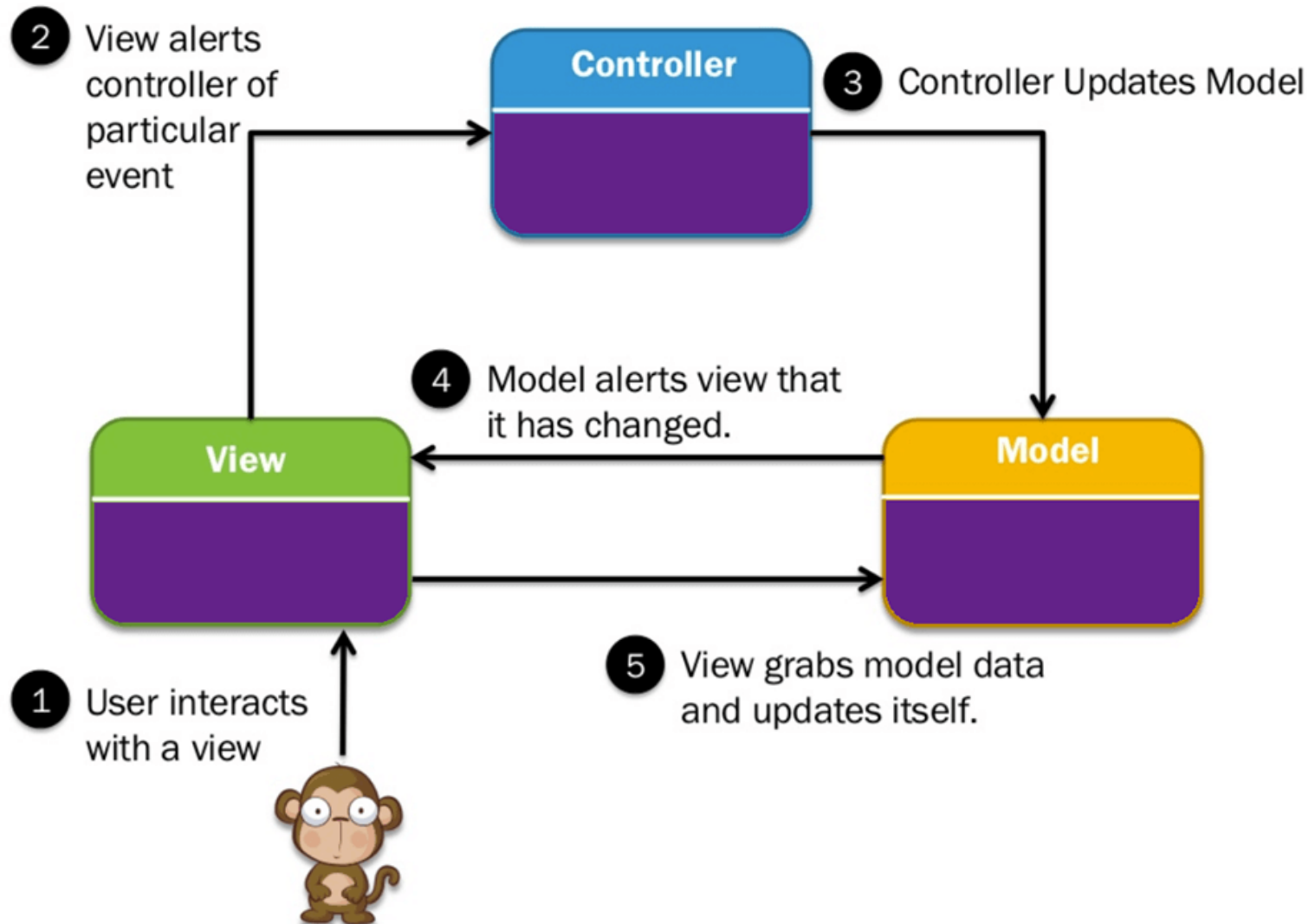


Tentukan pilihan kalian??? Kalian mau jadi seperti apa? Hanya sekedar tahu?
Atau mau jadi ahli (*expert*) pada bidang tertentu?

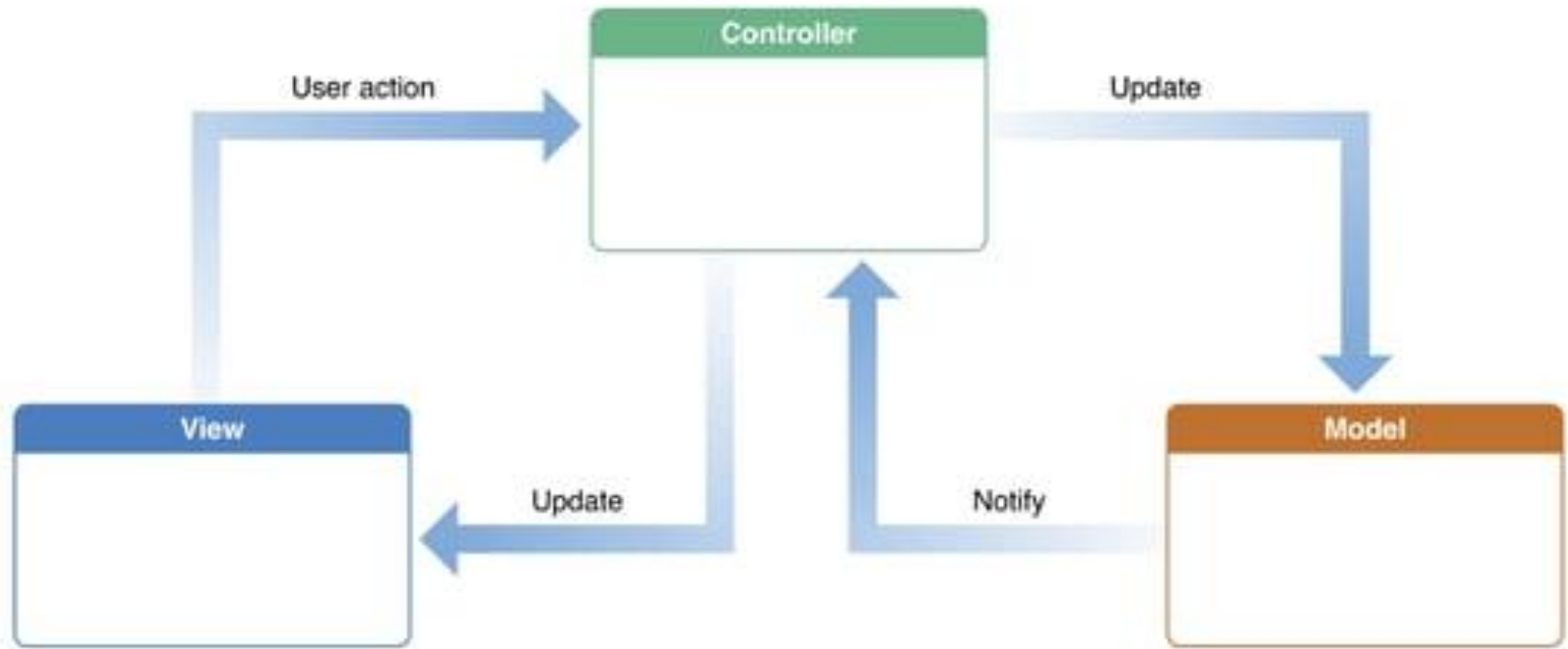
What is MVC Framework?

- ❑ The Model-View-Controller (MVC) framework is an architectural pattern that separates an application into *three main logical components* Model, View, and Controller.
- ❑ Each architecture component *is built to handle specific development aspect* of an application.
- ❑ MVC separates the business logic and presentation layer from each other.
- ❑ It was traditionally used for **desktop graphical user interfaces** (GUIs).
- ❑ Nowadays, MVC architecture has become popular for designing **web applications** as well as **mobile apps**.

MVC Architecture Diagram



MVC *with* User Action



The pictures above show what happens in a MVC web app when a user clicks a button, from the perspective of the user.

Why choose PHP?



Why choose

PHP

programming language
for web development?



Why choose PHP?

Usage statistics of PHP for websites

Request an **extensive PHP market report**.

[Learn more](#)

<https://w3techs.com/technologies/details/pl-php>

These diagrams show the usage statistics of PHP as server-side programming language on the web. See [technologies overview](#) for explanations on the methodologies used in our surveys. Our reports are updated daily.

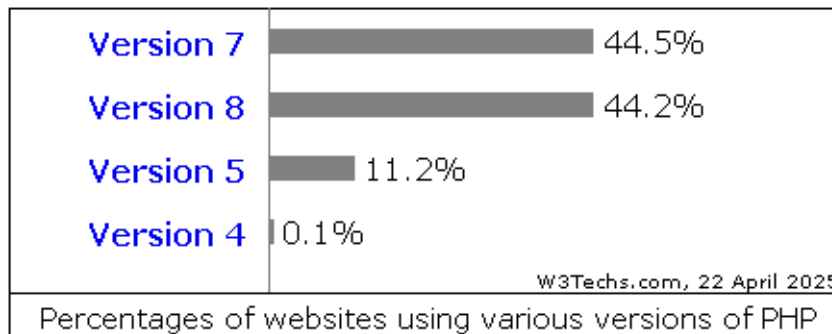
PHP is used by 74.3% of all the websites whose server-side programming language we know.

Versions of PHP

This diagram shows the percentages of websites using various versions of PHP.

How to read the diagram:

Version 7 is used by 44.5% of all the websites who use PHP.



Popular sites using PHP

- [Microsoft.com](#)
- [Facebook.com](#)
- [Wikipedia.org](#)
- [Wordpress.org](#)
- [Vimeo.com](#)
- [Digicert.com](#)
- [Wordpress.com](#)
- [Nih.gov](#)
- [Forbes.com](#)
- [Mozilla.org](#)

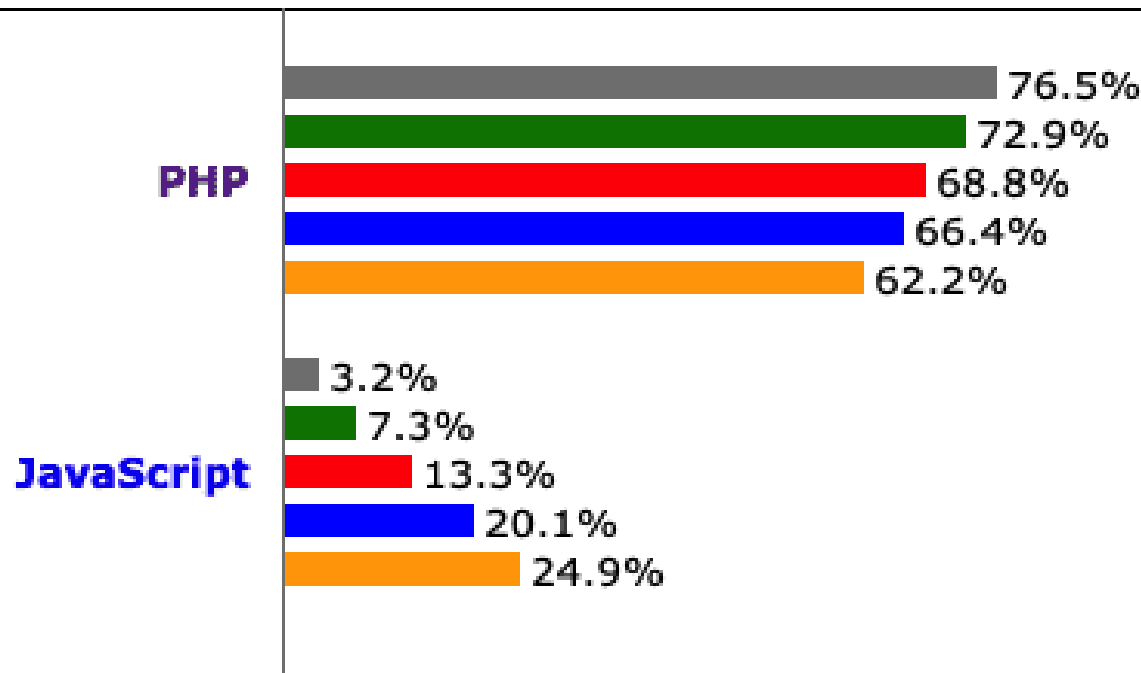
Random selection of sites using PHP

- [Deine-tierwelt.de](#)
- [Edinos.pl](#)
- [Erclyes.edu.tr](#)
- [Extra.ie](#)
- [Theosgyrosseattle.com](#)

PHP is Popular

According to **W3Techs**, PHP is used by around **74.3%** of all websites. It's eight times more popular than ASP.NET, its nearest rival in server-side programming languages.

Is PHP still relevant in 2024?



W3Techs.com, 6 March 2024

■ Overall ■ top 1,000,000 ■ top 100,000 ■ top 10,000 ■ top 1,000

Percentages of websites using the selected server-side programming languages broken down by ranking

How many websites are still powered by PHP in 2024?

If you have any interest in PHP, you might have heard the famous statistic that PHP fuels 80% of the whole Internet. But how true is it as of today? According to the latest metrics from **W3Techs**, in 2024, PHP still powers 76.5% of websites where the server-side programming language can be discerned.

PHP Development Companies

World Class Web Development Services



PHP is a **server side scripting language**. That is used to develop Static websites or Dynamic websites or Modern Web applications.

A decorative border of green leaves and branches at the top of the slide.

2 - What is PHP?

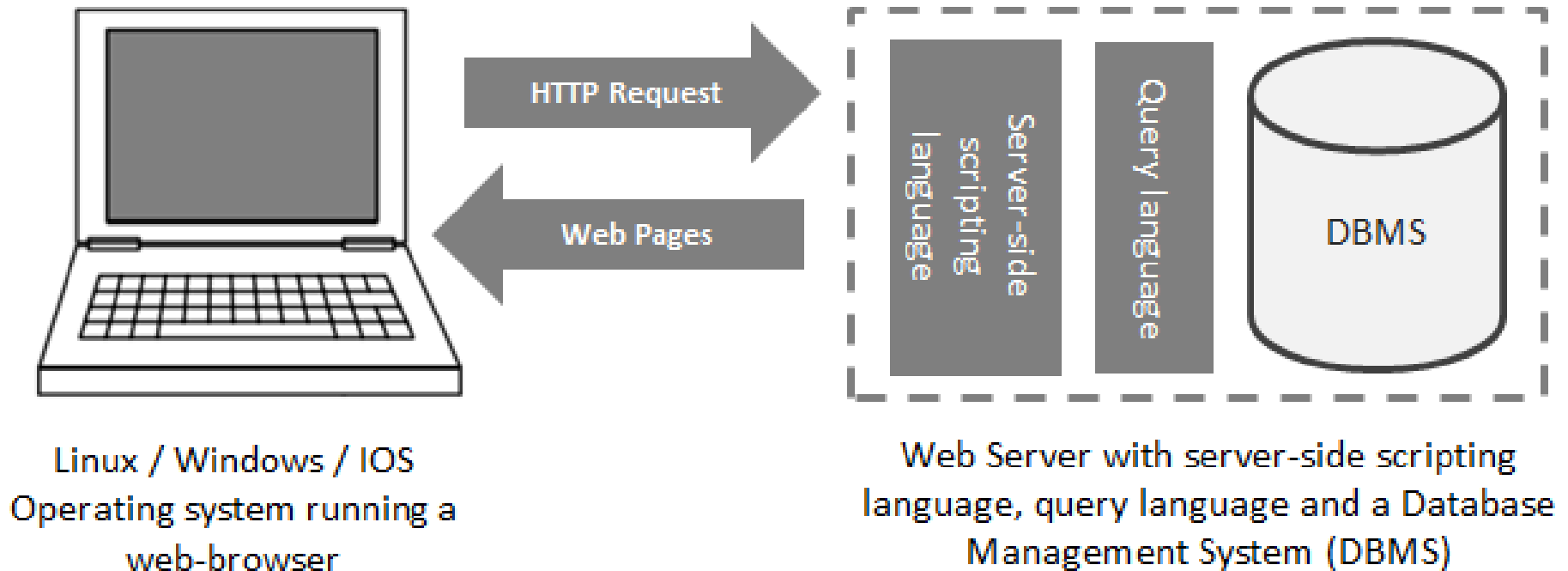
Write your first PHP Program

A decorative border of green leaves and branches at the bottom of the slide.


```
57 ga.type = 'text/javascript';
58 ga.async = true;
59 ga.src = ('https:' == document.location.protocol ? 'https://ssl' : 'http://www') +
60 'google-analytics.com/ga.js';
61 var s = document.getElementsByTagName('script')[0];
62 s.parentNode.insertBefore(ga, s);
63 }});
64 </script>
65 <?php
66 if (is_singular() && get_option('thread_comments')) {
67     wp_enqueue_script('comment-reply');
68 }
69 ?>
70 <?php wp_head(); ?>
71 </head>
72 <body <?php body_class(); ?>>
73 <div id="header">
74 <div class="wrapper">
75 <h1>
76 <?php if (is_front_page() && Spaged < 2): ?>
77 " />
78 <?php else : ?>
79 " />
80 </div>
81 </div>
82 </body>
83 </html>
```

Menurut Anda, apakah HTML adalah sebuah programming language?

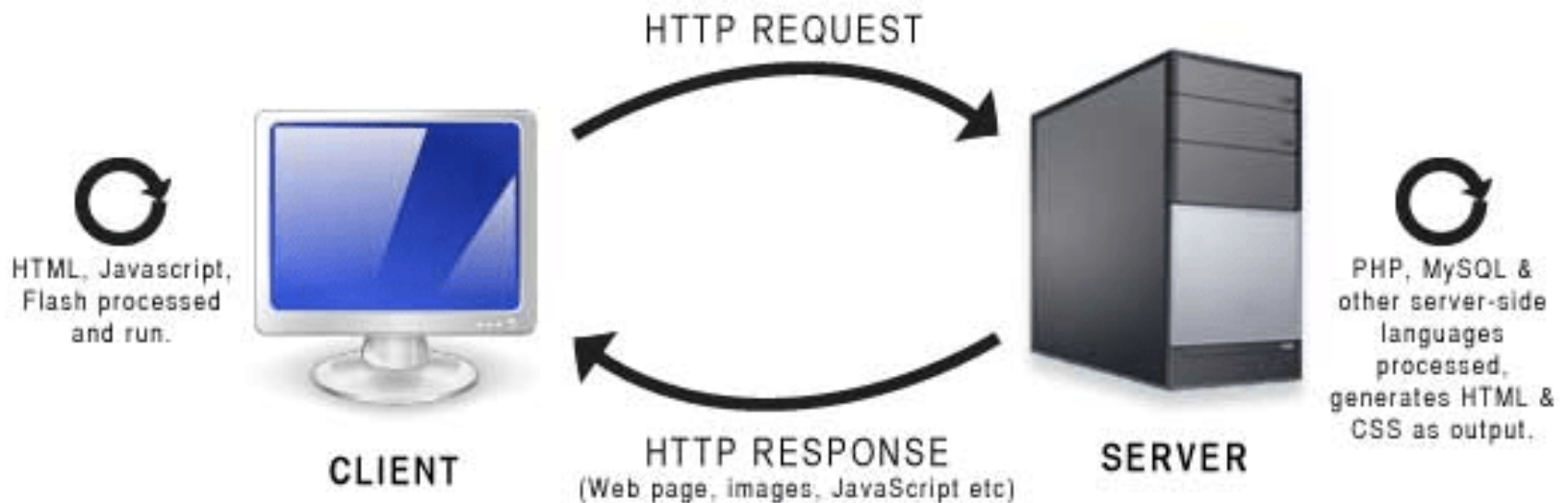
Server Side Technologies



The term '**server-side technologies**' can include/encompass a range of software solutions, such as **server-side scripting languages**; **Database Management Systems** (DBMS); **web server** software such as **Apache**; and many other technologies depending upon the application being built.

Server-side Scripting

Server-side scripts are executed on the server before the content is sent to the client's web browser.



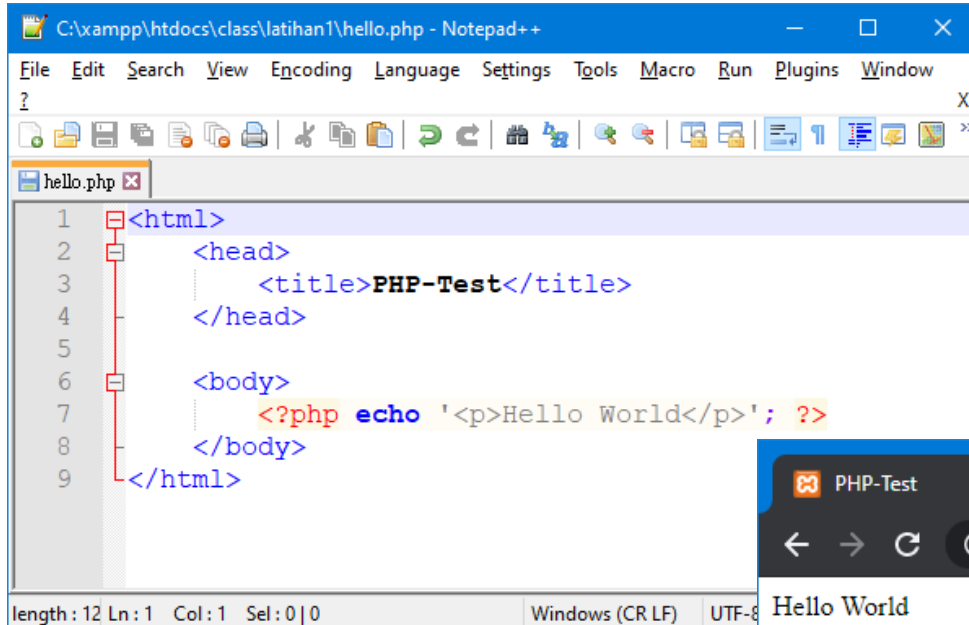
Server-side scripting languages are used to generate **dynamic web pages**, interact with **databases**, **handle user authentication**, and **perform other tasks that require server-side processing**. A server-side scripting language is a programming language used to create scripts that run on the server to generate dynamic content for web pages. Examples of popular server-side scripting languages include PHP, Python, Ruby, Java, and Node. Js.

What is PHP?

- ❖ PHP stands for **PHP: Hypertext Preprocessor**.
- ❖ PHP is a widely-used, **open source scripting language**.
- ❖ PHP scripts are **executed on the server**.
Apakah ada script yang di execute di sisi client/browser? Perbedaaannya apa?
- ❖ PHP is free to download and use.

Example PHP Script

Your Php Code

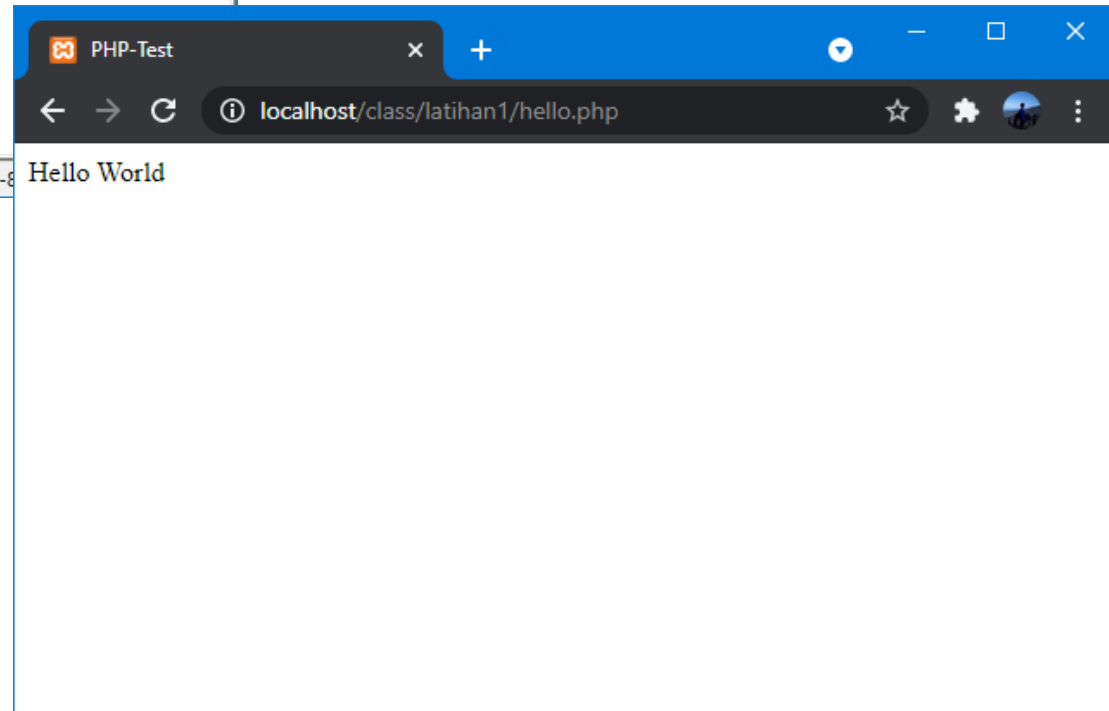


The screenshot shows the Notepad++ editor with the file 'hello.php' open. The code is as follows:

```
1 <html>
2   <head>
3     <title>PHP-Test</title>
4   </head>
5
6   <body>
7     <?php echo '<p>Hello World</p>'; ?>
8   </body>
9 </html>
```

The status bar at the bottom indicates 'length: 12 Ln: 1 Col: 1 Sel: 0 | 0', 'Windows (CR LF)', and 'UTF-8'.

**Access from
your web
browser**



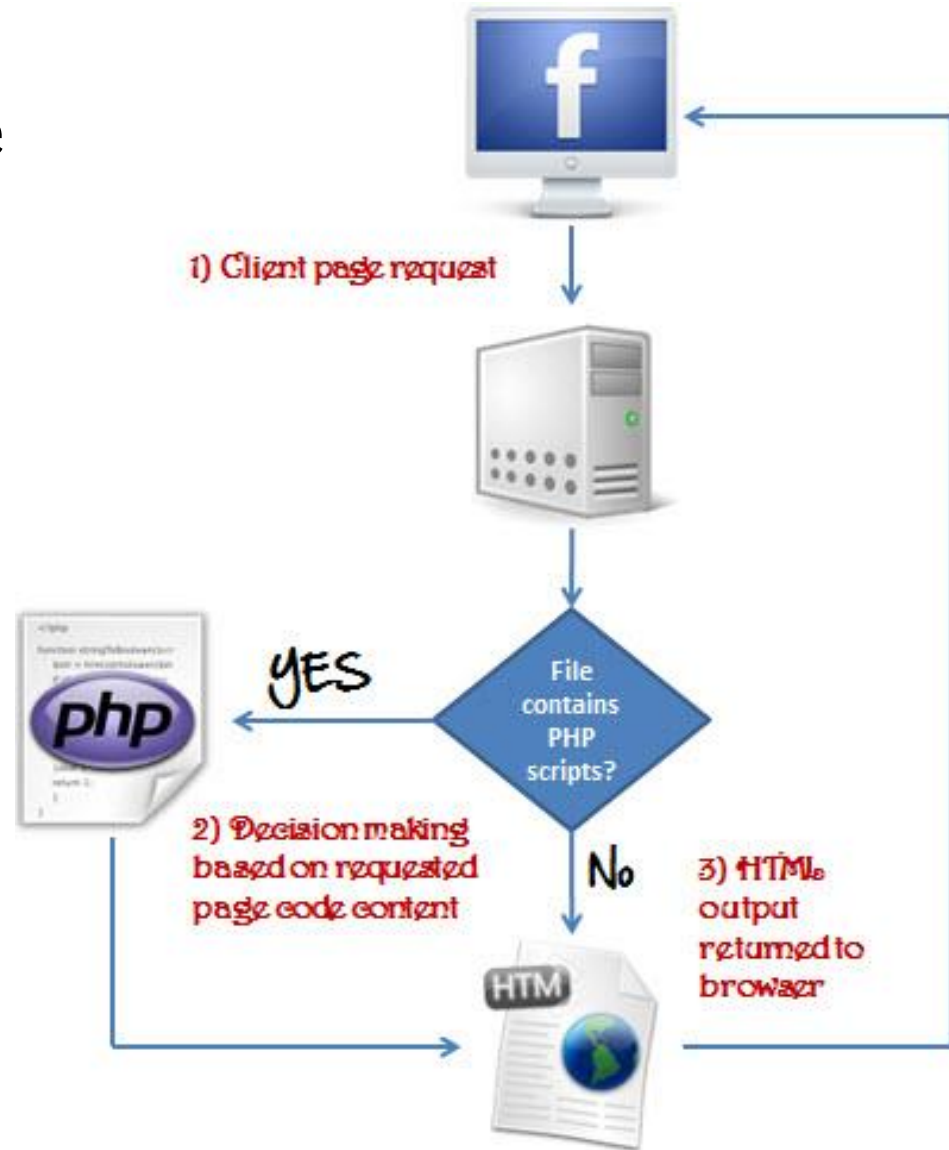


Pastikan semua PC sudah terinstall Xampp !!!

Demo in Class

Basic Architecture of PHP

The flowchart diagram shown below illustrates the basic architecture of a PHP web application and ***how the server handles the requests.***



Various Server Side Scripting Languages

Comparison the various server side scripting languages with PHP.

FEATURE	PHP	ASP	JSP	CFML
Learning curve	short	Longer than PHP	Longer than PHP	Longer than PHP
Web hosting	Supported by almost all hosting servers	Needs dedicated server	Fairly supported	Needs dedicated server
Open source	Yes	No	Yes	Both commercial and open source
Web services support	Built in	Uses the .NET framework	Uses add on libraries	Built in
Integration with HTML	Easy	Fairly complex	Fairly complex	Easy
MySQL support	Native	Needs third party drivers	Needs third party drivers	Current version has native support. Older versions use ODBC
Easily extended by other languages	Yes	No	Extended using Java classes and libraries.	Yes

ASP – Active Server Pages,
JSP – Java Server Pages,
CFML – Cold Fusion Markup language

What is a PHP File?

- ❖ PHP files can contain text, HTML, JavaScript code, and PHP code.
- ❖ PHP code are executed on the server, and the result is returned to the browser as plain HTML.
- ❖ PHP files have a default file extension of *".php"*.

What Can PHP Do?

- ❖ PHP can **generate dynamic page** content.
- ❖ PHP can **create, open, read, write, and close files** on the server.
- ❖ PHP can **collect form data (POST and GET)**.
- ❖ PHP can send and **receive cookies**.
- ❖ PHP can **add, delete, modify data in the database**.
- ❖ PHP can **restrict users to access** some pages on your website.
- ❖ PHP can **encrypt data**.

Why PHP?

- ❖ PHP runs on different platforms (Windows, Linux, Unix, Mac OS X, etc.)
- ❖ PHP is compatible with almost all servers used today (Apache, IIS, etc.)
- ❖ PHP has support for a wide range of databases
- ❖ PHP is free. Download it from the official PHP resource: www.php.net
- ❖ PHP is easy to learn and runs efficiently on the server side.

Set Up PHP *on* Your Own PC

- However, if your server does not support PHP, you must:
 - *install a web server*
 - *install PHP*
 - *install a database, such as MySQL*
- The official PHP website (PHP.net) has installation instructions for
PHP: <http://php.net/manual/en/install.php>

Basic PHP Syntax

- ❑ A PHP script can be placed anywhere in the document.
- ❑ A PHP script starts with **<?php** and ends with **?>**.

Example PHP Script:

```
<?php  
    // PHP code goes here  
?>
```

The default file extension for PHP files is ".php".

Comments *in* PHP

□ Example HTML Tag and PHP

<html>

<body>

```
<?php
```

```
    // This is a PHP comment line
```

```
    /*
```

```
        This is a PHP comment block
```

```
    */
```

```
    echo "Hello all students :)";
```

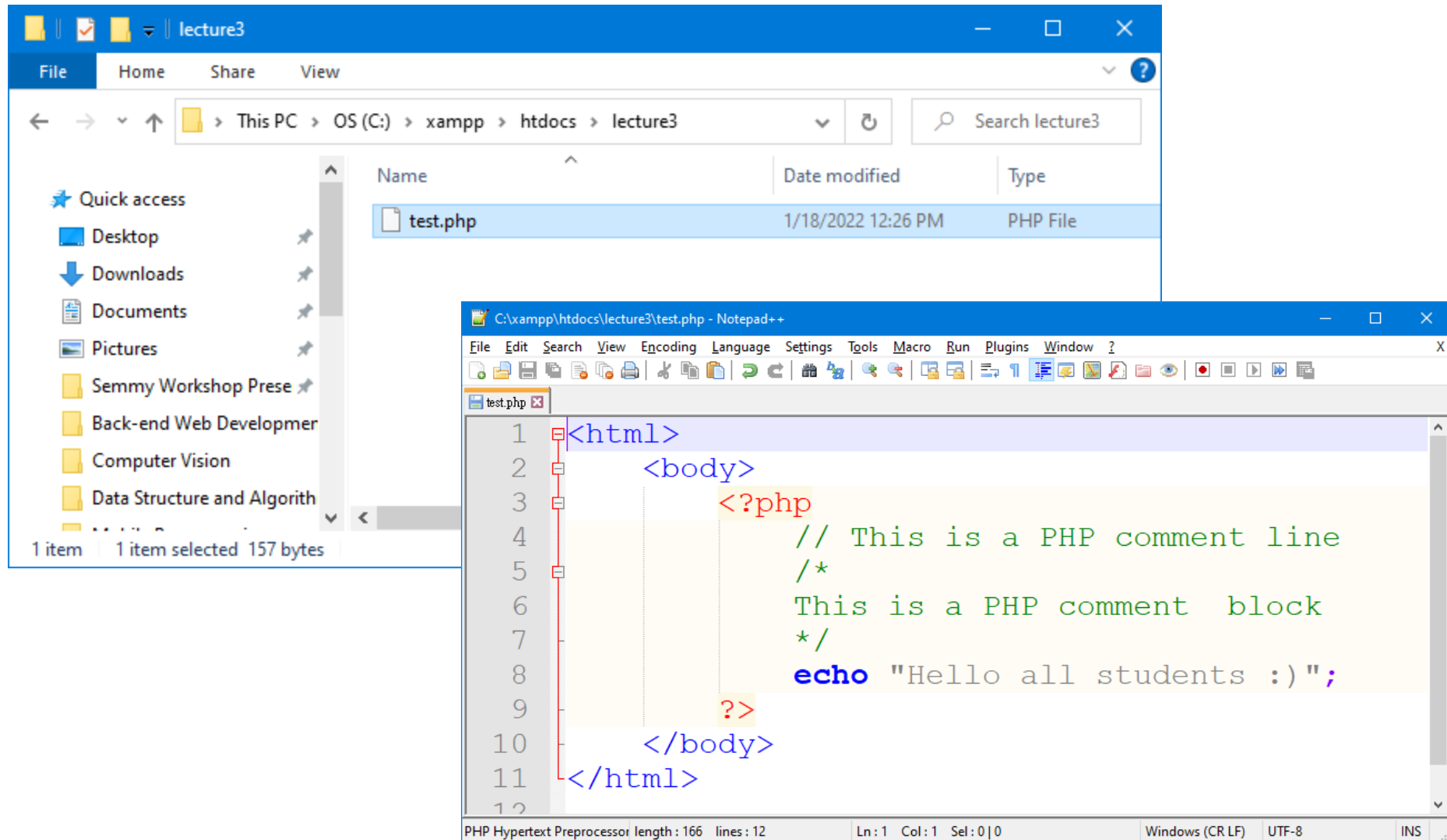
```
?>
```

</body>

</html>

HTML Tag & PHP

Create a folder in **htdocs** directory and create a new PHP file.



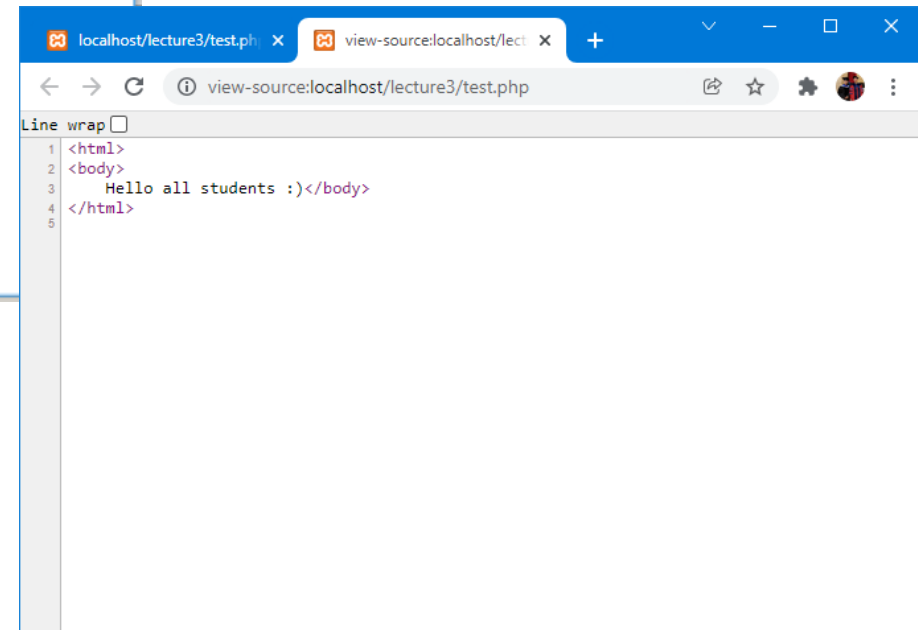
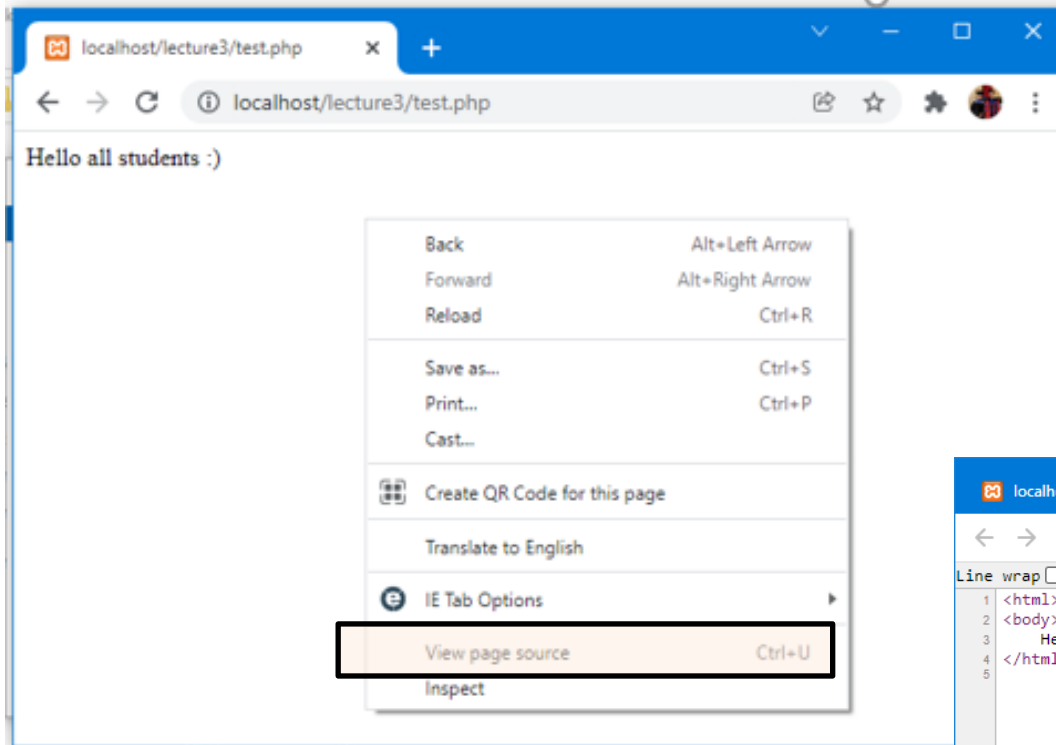
The image displays two windows from a Windows operating system. The top window is a File Explorer window titled 'lecture3', showing the directory path 'This PC > OS (C:) > xampp > htdocs > lecture3'. A file named 'test.php' is listed with a date modified of '1/18/2022 12:26 PM' and a type of 'PHP File'. The bottom window is a Notepad++ editor titled 'C:\xampp\htdocs\lecture3\test.php - Notepad++'. The editor shows the following code:

```
1 <html>
2   <body>
3     <?php
4       // This is a PHP comment line
5       /*
6       This is a PHP comment block
7       */
8       echo "Hello all students :)";
9     ?>
10  </body>
11 </html>
```

The status bar at the bottom of the Notepad++ window indicates 'PHP Hypertext Preprocessor length : 166 lines : 12', 'Ln : 1 Col : 1 Sel : 0 | 0', 'Windows (CR LF)', 'UTF-8', and 'INS'.

View Page Source

Right click in browser and click “view page source”



Mengapa “**View Page Source**” Tidak Menampilkan Kode PHP ?

Tidak Menampilkan Kode PHP di Browser ?

Server side Scripts *vs* Client side Script

Karena PHP kode hanya di instruksi untuk server web. Bahasa pemrograman pada sisi server lainnya termasuk C #, Python, Ruby, C ++ dan Java.

PHP Variables

- Variable can have short names (*like x and y*) or more descriptive names (*age, carname, totalvolume*).
- **Rules for PHP variables:**
 - *A variable starts with the \$ sign, followed by the name of the variable.*
 - A variable name must begin with a letter or the underscore character
 - A variable name can only contain alpha-numeric characters and underscores (A-z, 0-9, and _)
 - A variable name should not contain spaces
 - Variable names are case sensitive (\$y and \$Y are two different variables)

Simbol dollar(\$)
merupakan simbol
variabel dalam PHP.

Creating (Declaring) PHP Variables

- PHP has no command for declaring a variable.
- A variable is created the moment you first assign a value to it:
- ```
$txt="Hello world!";
$x=5;
```
- After the execution of the statements above, the variable **txt** will hold the value **Hello world!**, and the variable **x** will hold the value **5**.
- **Note:** When you assign a text value to a variable, put quotes around the value.

# PHP is a Loosely Typed Language

- In the example above (*previous slide*), notice that we did not have to tell PHP which data type the variable is.
- **PHP automatically converts the variable to the correct data type, depending on its value.**
- In a strongly typed programming language, we will have to declare (***define***) the ***type*** and ***name*** of the variable before using it.

# PHP Variable Scopes

- The scope of a variable is the part of the script where the variable can be referenced/used.
- PHP has four different variable scopes:
  - *local*
  - *global*
  - *static*
  - *parameter*

# 1 - Local Scope

- A variable declared **within** a PHP function is **local** and can only be accessed within that function:

- **Example**

```
<?php
 $x=5; // global scope

 function myTest()
 {
 echo $x; // local scope
 }

 myTest();

?>
```

---

## 2 - Global Scope

- A variable that is defined outside of any function, has a global scope.
- Global variables can be accessed from any part of the script, EXCEPT from within a function.
- To access a global variable from within a function, use the ***global*** keyword:

# Global Scope

## □ Example

```
<?php
 $x=5; // global scope
 $y=10; // global scope

 function myTest()
 {
 global $x,$y;
 $y=$x+$y;
 }
 myTest();
 echo $y; // outputs 15
?>
```



## 3 - Static Scope

- *When a function is completed, all of its variables are normally deleted. However, sometimes you want a local variable to not be deleted.*
- To do this, use the **static** keyword when you first declare the variable:

# Static Scope

## □ Example

```
<?php
 function myTest()
 {
 static $x=0;
 echo $x;
 $x++;
 }

 myTest();
 myTest();
 myTest();

?>
```

## □ OUTPUT ?

# Static Scope

The screenshot shows the Paiza.io Online PHP Editor interface. The browser address bar displays the URL `paiza.io/projects/gAH1KEAmeY5vKM1gB_vgOg`. The editor's header includes the Paiza.io logo, navigation links for "New code", "Recent code", and "WebDev", and buttons for "Sign Up" and "Sign In". A "Beta" badge is visible on the left. The editor is set to "PHP" and has a tab for "Main.php". A green success banner at the top of the code editor area reads "Success" with "Tweet" and "Share 0" options. The code editor contains the following PHP code:

```
1 <?php
2 function myTest(){
3 static $x=0;
4 echo $x;
5 $x++;
6 }
7 myTest();
8 myTest();
9 myTest();
10
11 ?>
12
```

Below the code editor is a toolbar with a "Run (Ctrl-Enter)" button and a "PHP books" link. To the right of the toolbar are several icons for user profile, information, settings, clock, undo, star, group, lock, and share. The bottom section of the interface shows the "Output" tab with the text "012" and a "(0.03 sec)" execution time indicator. A "Text" dropdown menu is also present. The PaizaCloud logo is located in the bottom right corner.

## 4- Parameter Scope

- *A parameter is a local variable whose value is passed to the function by the calling code.*
- Parameters are declared in a parameter list as part of the function declaration:

□ Example :

```
<?php
 function myTest($x)
 {
 echo $x;
 }
 myTest(5);
?>
```

# PHP **echo** & **print** Statements

- ❑ In PHP there is two basic ways to get output: **echo** and **print**.
- ❑ There are some differences between **echo** and **print**:
  - ❑ **echo** - can output one or more strings
  - ❑ **print** - can only output one string, and returns always 1
- ❑ **Tip:** **echo** is marginally faster compared to **print** as **echo** does not return any value.

```
<?php
 echo "<h2>PHP is fun!</h2>";
 echo "Hello world!
";
 echo "This", " string", " was", " made", " with multiple
 parameters.";
?>
```

# PHP Arrays

- An array stores multiple values in one single variable.

## Example:

```
<?php
$cars = array("Volvo","BMW","Toyota");
echo "I like " . $cars[0] . ", " . $cars[1] . " and " . $cars[2] . ".";
?>
```

# Create an Array in PHP

- In PHP, the array() function is used to create an array:
- **Example :**  

```
$cars = array("Volvo","BMW","Toyota");
```
- In PHP, there are *three types* of arrays:
  - 1) **Indexed arrays** - Arrays with numeric index
  - 2) **Associative arrays** - Arrays with named keys
  - 3) **Multidimensional arrays** - Arrays containing one or more arrays.

# 1 - PHP Indexed Arrays

- There are two ways to create indexed arrays:
  - The index can be assigned automatically (index always starts at 0):
  - `$cars=array("Volvo","BMW","Toyota");`

**OR**

- The index can be assigned manually:  
`$cars[0]="Volvo";`  
`$cars[1]="BMW";`  
`$cars[2]="Toyota";`



## 2 - PHP Associative Arrays

- Associative arrays are arrays that use **named keys** that you assign to them.
- There are two ways to create an associative array:
  - `$age=array("Peter"=>"35","Ben"=>"37","Joe"=>"43 ");`

OR

- `$age['Peter']="35";  
$age['Ben']="37";  
$age['Joe']="43";`

## 2 - PHP Associative Arrays

### □ Example:

```
<?php
 $age=array("Peter"=>"35","Ben"=>"37","Joe"=>"43");
 echo "Peter is " . $age['Peter'] . " years old.";
?>
```

**// Output:**

*Peter is 35 years old.*

# 3- PHP Multidimensional Arrays

- An array can also ***contain another array as a value***, which in turn can hold other arrays as well. In such a way we can create *two- or three-dimensional arrays*:

- **Example:**

```
<?php
// A two-dimensional array:
$cars = array (
 array("Volvo",100,96),
 array("BMW",60,59),
 array("Toyota",110,100)
);
?>
```

# The foreach Loop

The foreach loop works only on arrays, and is used to loop through each key/value pair in an array.

```
foreach ($array as $value)
{
 code to be executed;
}
```

For every loop iteration, the value of the current array element is assigned to `$value` (*and the array pointer is moved by one*) - so on the next loop iteration, you'll be looking at the next array value.

# The foreach Loop

## Example:

```
<?php
 $x=array("one","two","three");
 foreach ($x as $value)
 {
 echo $value . "
";
 }
?>
```

# PHP Functions

## □ PHP User Defined Functions

- Besides the built-in PHP functions, we can create our own functions.
- A function is a block of statements that can be used repeatedly in a program.
- A function will not execute immediately when a page loads.
- A function will be executed by a call to the function.

## □ Create a User Defined Function in PHP

- A user defined function declaration starts with the word "function":

### Syntax

```
function functionName(arg1, arg2...)
{
 code to be executed;
}
```

- **Note:** A function name can start with a letter or underscore (not a number).

# GET vs POST

|                             | GET                                                                                                                                                  | POST                                                                                                           |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| BACK button/Reload          | Harmless                                                                                                                                             | Data will be re-submitted (the browser should alert the user that the data are about to be re-submitted)       |
| Bookmarked                  | Can be bookmarked                                                                                                                                    | Cannot be bookmarked                                                                                           |
| Cached                      | Can be cached                                                                                                                                        | Not cached                                                                                                     |
| Encoding type               | application/x-www-form-urlencoded                                                                                                                    | application/x-www-form-urlencoded or multipart/form-data. Use multipart encoding for binary data               |
| History                     | Parameters remain in browser history                                                                                                                 | Parameters are not saved in browser history                                                                    |
| Restrictions on data length | Yes, when sending data, the GET method adds the data to the URL; and the length of a URL is limited (maximum URL length is 2048 characters)          | No restrictions                                                                                                |
| Restrictions on data type   | Only ASCII characters allowed                                                                                                                        | No restrictions. Binary data is also allowed                                                                   |
| Security                    | GET is less secure compared to POST because data sent is part of the URL<br><br>Never use GET when sending passwords or other sensitive information! | POST is a little safer than GET because the parameters are not stored in browser history or in web server logs |
| Visibility                  | Data is visible to everyone in the URL                                                                                                               | Data is not displayed in the URL                                                                               |

# PHP \$\_GET Variable

- The predefined \$\_GET variable is used to collect values in a form with method="get"
- Information sent from a form with the GET method is visible to everyone (it will be displayed in the browser's address bar) and has limits on the amount of information to send.

- Example

```
<form action="welcome.php" method="get">
 Name: <input type="text" name="fname">
 Age: <input type="text" name="age">
 <input type="submit">
</form>
```



# Example ⇒ PHP \$\_GET Variable

- When the user clicks the "Submit" button, the URL sent to the server could look something like this:  
<http://www.w3schools.com/welcome.php?fname=Peter&age=37>
- The "welcome.php" file can now use the \$\_GET variable to collect form data as follow:

```
Welcome <?php echo $_GET["fname"]; ?>.
 You
are <?php echo $_GET["age"]; ?> years old!
```

# The \$\_POST Variable

- ❑ The predefined **\$\_POST** variable is used to collect values from a form sent with method="post".
- ❑ Information sent from a form with the POST method is invisible to others and has no limits on the amount of information to send.
- ❑ **Note:** However, there is an 8 MB max size for the POST method, by default (can be changed by setting the post\_max\_size in the *php.ini* file).



# Exercises *for* Students

# 1 - Membuat Variabel *di* PHP

**Create directory** : *C:/xampp/htdocs/class3/*

**Create file** : latihan1.php

```
<?php

// membuat variabel baru
$stok = 40;

// mengisi ulang variabel dengan nilai baru
$stok = 34;
```

Tampilkan output di browser !!!

## 2 - Mengambil Nilai dari Variabel

**Create directory** : *C:/xampp/htdocs/class3/*  
**Create file** : latihan2.php

```
<?php

// membuat variabel baru
$nama_barang = "Minyak Goreng";
$harga = 15000;

// menampilkan isi variabel
echo "Ibu membeli $nama_barang seharga Rp $harga";
```

Jika kita menggunakan tanda petik ganda (") dalam echo, maka kita bisa menuliskan langsung nama variabelnya.

```
$judul = "Belajar PHP dari nol sampai expert";
echo "Judul artikel: $judul";
```

Apabila kita menggunakan tanda petik tunggal ('), maka kita harus menggunakan titik untuk menggabungkan teks dengan variabelnya.

```
$judul = "Tutorial PHP untuk Pemula";
echo 'Judul artikel: '.$judul;
```

# 3 - Mengenal Tipe Data di PHP

**Create directory** : class3

**Create file** : latihan3.php

```
<?php

// tipe data char (karakter)
$jenis_kelamin = 'L';

// tipe data string (teks)
$nama_lengkap = "Petani Kode";

// tipe data integer
$umur = 20;

// tipe data float
$berat = 48.3;

// tipe data float
$tinggi = 182.2;

// tipe data boolean
$menikah = false;

echo "Nama: $nama_lengkap
";
echo "Jenis Kelamin: $jenis_kelamin
";
echo "Umur: $umur tahun
";
echo "berat badan: $berat kg
";
echo "tinggi badan: $tinggi cm
";
echo "menikah: $menikah";
```

Ada beberapa macam tipe data yang dapat disimpan dalam variabel:

1. Tipe data char (karakter)
2. Tipe data string (teks)
3. Tipe data integer (angka)
4. Tipe data float (pecahan)
5. Tipe data boolean
6. Tipe data objek
7. Tipe data Array
8. NULL
9. dll.

## Tipe Data Array dalam PHP

```
$minuman = array("Kopi", "Teh", "Jus Jeruk");
$makanan = ["Nasi Goreng", "Soto", "Bubur"];
```

# 4 - Konversi Tipe data di PHP

**Create directory** : class3

**Create file** : latihan4.php (*digabung dua file dibawah*)

Pada PHP konversi tipe data bisa dilakukan dengan operator kali (\*)

```
<?php
$foo = "1"; // mula-mula $foo dalam bentuk string (ASCII 49)
$foo *= 2; // $foo sekarang adalah integer (2)
$foo = $foo * 1.3; // $foo sekarang adalah float (2.6)
$foo = 5 * "10 Little Piggies"; // $foo sekarang adalah integer (50)
$foo = 5 * "10 Small Pigs"; // $foo sekarang adalah integer (50)
?>
```

Bisa juga dilakukan seperti pada bahasa pemrograman java atau C.

```
<?php

$a = "32";
$a = (int) $a; // ubah nilai a menjadi integer
$a = (float) $a; // ubah nilai a menjadi float
$a = (string) $a; // ubah nilai a menjadi string

?>
```

# 5 - Menghapus Variabel dari Memori

**Create directory** : class3

**Create file** : latihan5.php

```
// membuat variabel $tmp
$tmp = 2901;

// menghapus variabel $tmp
unset($tmp);

// mencoba mengakses variabel $tmp
echo $tmp;
```

Cara menghapus variabel di PHP dapat menggunakan fungsi **unset()**.



# 6 - Operator Aritmatika

**Create directory** : class3

**Create file** : latihan6.php

```
<?php

$a = 5;
$b = 2;

// penjumlahan
$c = $a + $b;
echo "$a + $b = $c";
echo "<hr>";

// pengurangan
$c = $a - $b;
echo "$a - $b = $c";
echo "<hr>";

// Perkalian
$c = $a * $b;
echo "$a * $b = $c";
echo "<hr>";

// Pembagian
$c = $a / $b;
echo "$a / $b = $c";
echo "<hr>";

// Sisa bagi
$c = $a % $b;
echo "$a % $b = $c";
echo "<hr>";

// Pangkat
$c = $a ** $b;
echo "$a ** $b = $c";
echo "<hr>";

?>
```

Operator aritmatika merupakan operator untuk melakukan operasi aritmatika.

# 7 - Percabangan If/Elseif/Else

**Create directory** : class3

**Create file** : latihan7.php

```
<?php

$nilai = 88;

if ($nilai > 90) {
 $grade = "A+";
} elseif($nilai > 80){
 $grade = "A";
} elseif($nilai > 70){
 $grade = "B+";
} elseif($nilai > 60){
 $grade = "B";
} elseif($nilai > 50){
 $grade = "C+";
} elseif($nilai > 40){
 $grade = "C";
} elseif($nilai > 30){
 $grade = "D";
} elseif($nilai > 20){
 $grade = "E";
} else {
 $grade = "F";
}

echo "Nilai anda: $nilai
";
echo "Grade: $grade";

?>
```

# 8 - Percabangan Switch/Case

**Create directory** : class3

**Create file** : latihan8.php

```
<?php

$level = 3;

switch($level){
 case 1:
 echo "Pelajari HTML";
 break;
 case 2:
 echo "Pelajari CSS";
 break;
 case 3:
 echo "Pelajari Javascript";
 break;
 case 4:
 echo "Pelajari PHP";
 break;
 default:
 echo "Kamu bukan programmer!";
}

?>
```

# 9 - Percabangan Bersarang (*nested*)

**Create directory** : class3

**Create file** : latihan9.php

```
<?php

$umur = 17;
$menikah = false;

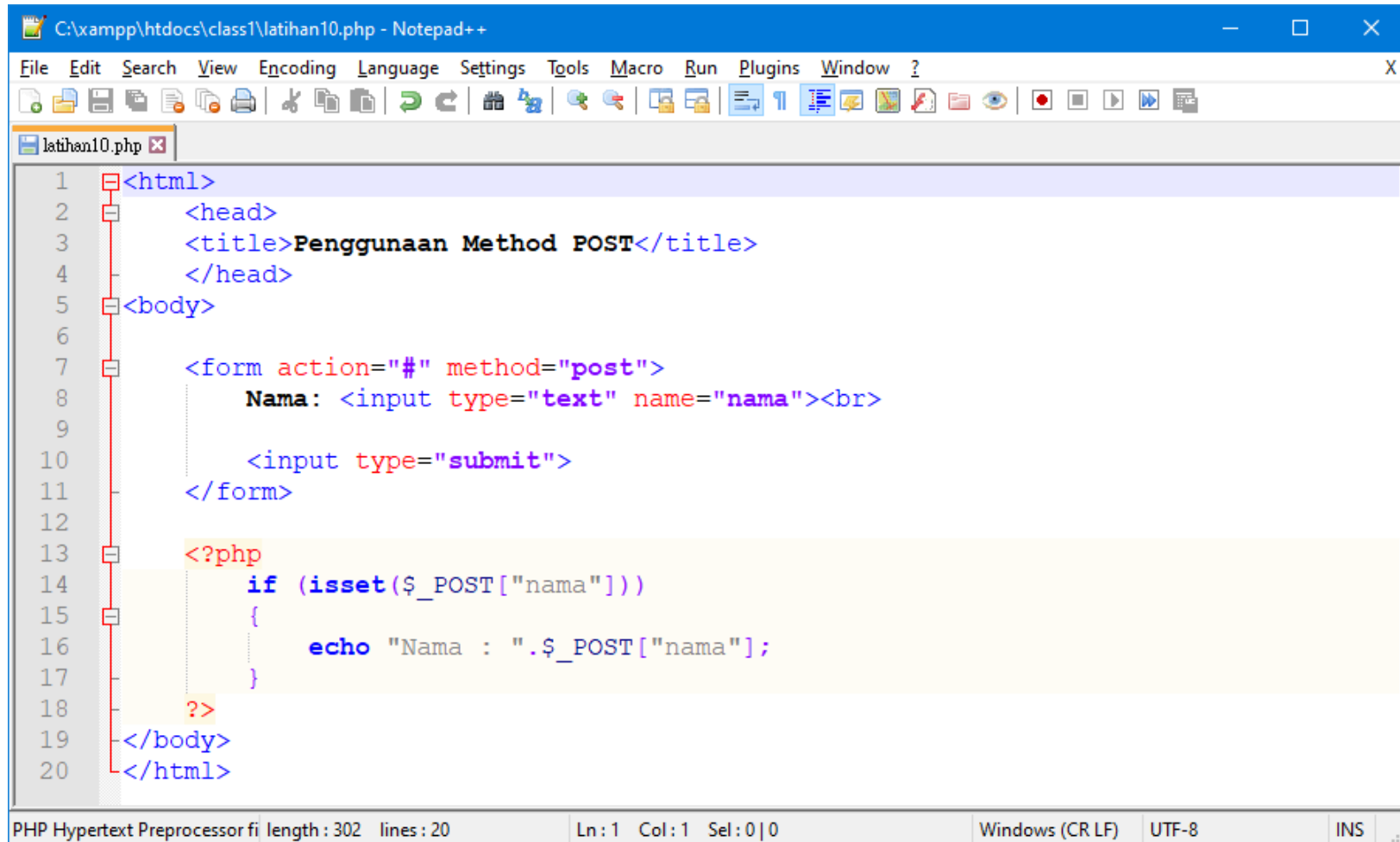
if($umur > 18){
 if($menikah){
 echo "Selamat datang pak!";
 } else {
 echo "Selamat datang wahai pemuda!";
 }
} else {
 echo "Maaf website ini hanya untuk yang sudah berumur 18+";
}

?>
```

# 10 - Penggunaan Method POST

**Create directory** : class3

**Create file** : latihan10.php



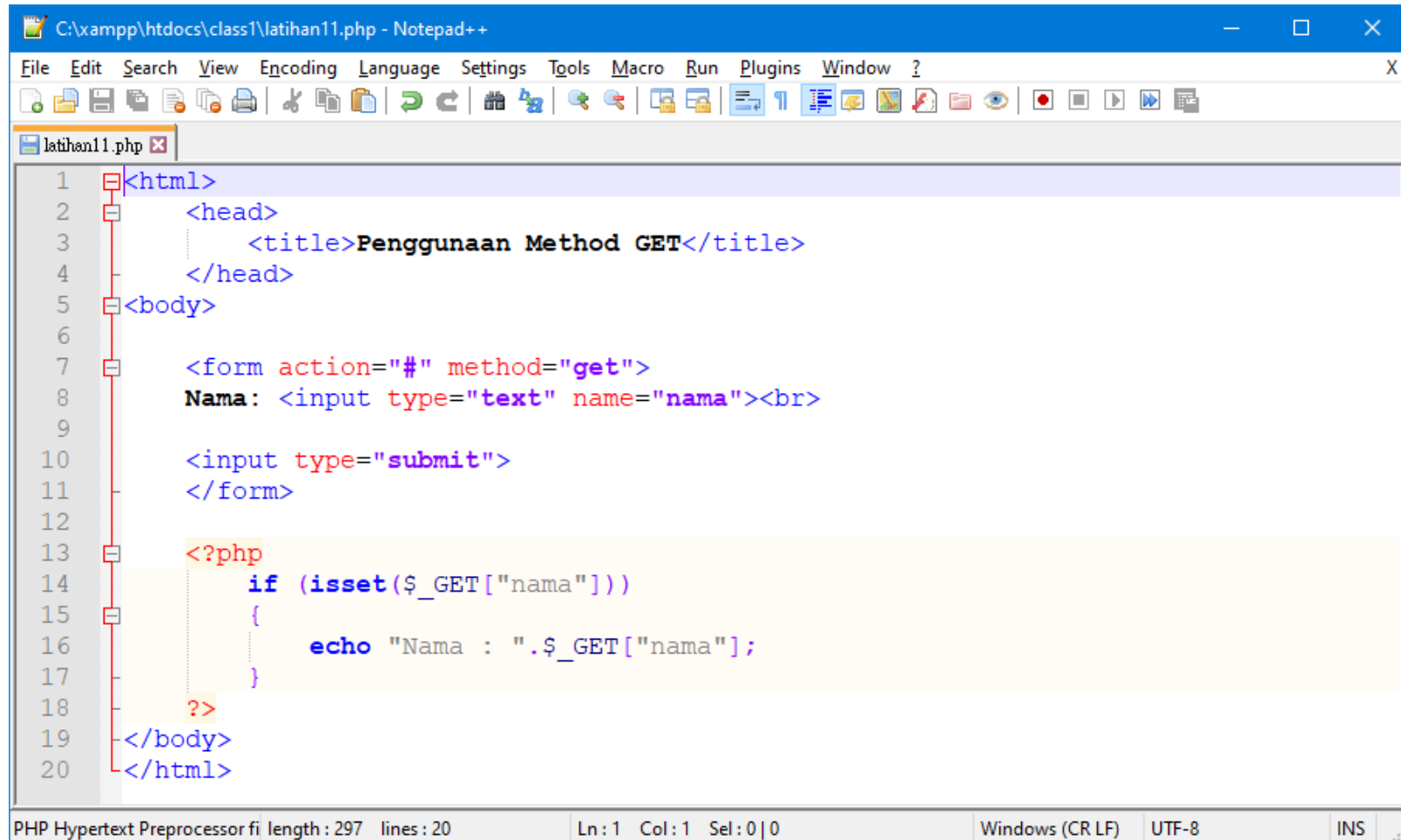
```
C:\xampp\htdocs\class1\latihan10.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
latihan10.php x
1 <html>
2 <head>
3 <title>Penggunaan Method POST</title>
4 </head>
5 <body>
6
7 <form action="#" method="post">
8 Nama: <input type="text" name="nama">

9
10 <input type="submit">
11 </form>
12
13 <?php
14 if (isset($_POST["nama"]))
15 {
16 echo "Nama : ".$_POST["nama"];
17 }
18 ?>
19 </body>
20 </html>
PHP Hypertext Preprocessor file length : 302 lines : 20
Ln : 1 Col : 1 Sel : 0 | 0 Windows (CR LF) UTF-8 INS
```

# 11 - Penggunaan Method GET

**Create directory** : class3

**Create file** : latihan11.php



```
C:\xampp\htdocs\class1\latihan11.php - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
latihan11.php
1 <html>
2 <head>
3 <title>Penggunaan Method GET</title>
4 </head>
5 <body>
6
7 <form action="#" method="get">
8 Nama: <input type="text" name="nama">

9
10 <input type="submit">
11 </form>
12
13 <?php
14 if (isset($_GET["nama"]))
15 {
16 echo "Nama : ".$_GET["nama"];
17 }
18 ?>
19 </body>
20 </html>
```

PHP Hypertext Preprocessor | length : 297 | lines : 20 | Ln : 1 Col : 1 Sel : 0 | 0 | Windows (CR LF) | UTF-8 | INS

# Perbedaan Method GET dan POST!

## Kapan sebaiknya digunakan ?

### 1. Contoh Penggunaan method POST



### 2. Contoh Penggunaan method GET



# Perbedaan Method GET dan POST!

## Kapan sebaiknya digunakan ?

### Kapan menggunakan method POST dan GET ?

#### #Method GET

- ❑ Jangan gunakan GET bila ingin membuat form untuk data yang sensitif/ mempunyai privasi, misalnya username, password (*karena akan tampil di URL*).
- ❑ Sebaliknya gunakan GET untuk data dengan informasi yang umum, seperti nama, email atau lainnya.
- ❑ Gunakan method GET untuk data yang relatif kecil.

#### #Method POST

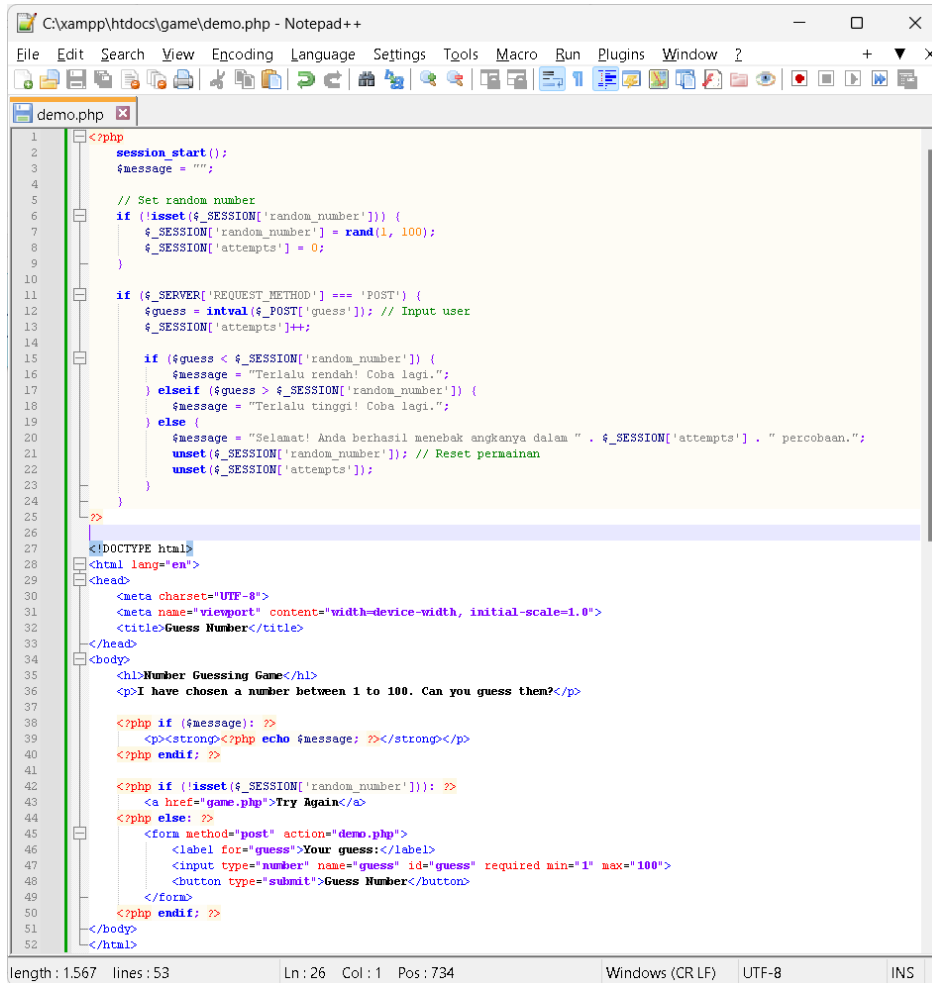
- ❑ Gunakan method POST untuk jenis informasi data pribadi misalnya username dan password, karena lebih aman dibandingkan method GET dan data tidak akan tampil di URL.
- ❑ Gunakan method POST untuk data yang relatif besar.



# 12 - Number Guessing Game

**Create directory** : game

**Create file** : C:/xampp/htdocs/game/demo.php



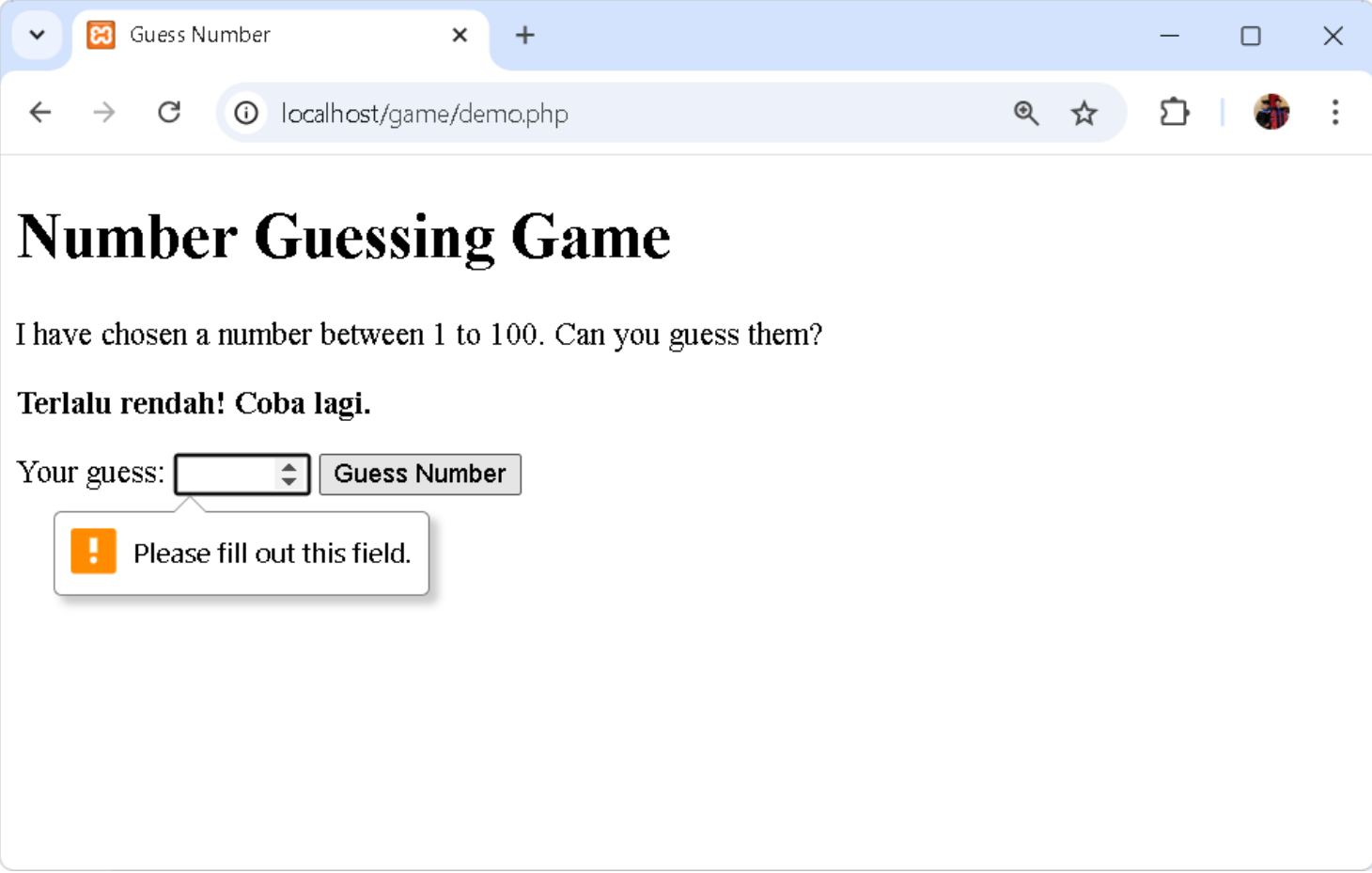
```
1 <?php
2 session_start();
3 $message = "";
4
5 // Set random number
6 if (!isset($_SESSION['random_number'])) {
7 $_SESSION['random_number'] = rand(1, 100);
8 $_SESSION['attempts'] = 0;
9 }
10
11 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
12 $guess = intval($_POST['guess']); // Input user
13 $_SESSION['attempts']++;
14
15 if ($guess < $_SESSION['random_number']) {
16 $message = "Terlalu rendah! Coba lagi.";
17 } elseif ($guess > $_SESSION['random_number']) {
18 $message = "Terlalu tinggi! Coba lagi.";
19 } else {
20 $message = "Selamat! Anda berhasil menebak angkanya dalam " . $_SESSION['attempts'] . " percobaan.";
21 unset($_SESSION['random_number']); // Reset permainan
22 unset($_SESSION['attempts']);
23 }
24 }
25
26 <DOCTYPE html>
27 <html lang="en">
28 <head>
29 <meta charset="UTF-8">
30 <meta name="viewport" content="width=device-width, initial-scale=1.0">
31 <title>Guess Number</title>
32 </head>
33 <body>
34 <h1>Number Guessing Game</h1>
35 <p>I have chosen a number between 1 to 100. Can you guess them?</p>
36
37 <?php if ($message): ?>
38 <p><?php echo $message; ?></p>
39 <?php endif: ?>
40
41 <?php if (!isset($_SESSION['random_number'])): ?>
42 Try Again
43 <?php else: ?>
44 <form method="post" action="demo.php">
45 <label for="guess">Your guess:</label>
46 <input type="number" name="guess" id="guess" required min="1" max="100">
47 <button type="submit">Guess Number</button>
48 </form>
49 <?php endif: ?>
50 </body>
51 </html>
```

Simpan file demo.php di folder server lokal, misalnya di htdocs (untuk XAMPP). Selanjutnya akses file melalui browser:  
<http://localhost/game/demo.php>.

# 12 - Number Guessing Game

**Create directory** : game

**Create file** : C:/xampp/htdocs/game/demo.php



The screenshot shows a web browser window with the title "Guess Number". The address bar displays "localhost/game/demo.php". The page content includes the title "Number Guessing Game", a message "I have chosen a number between 1 to 100. Can you guess them?", and a feedback message "Terlalu rendah! Coba lagi." (Too low! Try again.). Below this is a form with the label "Your guess:" followed by an empty text input field and a "Guess Number" button. A validation error message "Please fill out this field." is shown in a tooltip below the input field.

Guess Number

localhost/game/demo.php

## Number Guessing Game

I have chosen a number between 1 to 100. Can you guess them?

Terlalu rendah! Coba lagi.

Your guess:

! Please fill out this field.

Latihan 1-12 ini akan di submit  
*next meeting.*

Buat *report* dari screenshot (*code  
& output di browser*) per latihan  
dan akan dikumpulkan dalam  
format PDF (Google CR).



# Reference (#1)

## **MVC Architecture**

<https://www.javatpoint.com/php-mvc-architecture>

## **5 Hal Dasar yang Harus diketahui tentang Variabel dan Tipe Data**

<https://www.petanikode.com/php-variabel/>

## **7 Jenis Operator dalam PHP**

<https://www.petanikode.com/php-operator/>

## **Membuat Logika Program**

<https://www.petanikode.com/php-percabangan/>

## **MVC Tutorial for Beginners: What is, Architecture**

<https://www.guru99.com/mvc-tutorial.html>

## **MVC Architecture in 5 minutes: a tutorial for beginners**

<https://www.educative.io/blog/mvc-tutorial>

## **PHP MVC Framework Tutorial: CodeIgniter Example**

<https://www.guru99.com/php-mvc-frameworks.html>

# Reference (#2)

## **Pengertian Variabel dan Cara Penulisan Variabel PHP**

<https://www.duniailkom.com/pengertian-variabel-dan-cara-penulisan-variabel-php/>

## **PHP Introduction**

[https://www.w3schools.com/php/php\\_intro.asp](https://www.w3schools.com/php/php_intro.asp)

## **What is PHP?**

<https://www.php.net/manual/en/intro-what-is.php>

## **PHP Tutorial**

<https://www.tutorialspoint.com/php/index.htm>

## **What is PHP? Write your first PHP Program**

<https://www.guru99.com/what-is-php-first-php-program.html>

# END PRESENTATION

Thank you for your attention

Instructor: S – W – T

THANK YOU

